

Topic to discuss

- Gauss-Legendre Integration
- 2 point Gauss-Legendre
- 3 point Gauss-Legendre
- 4 point Gauss-Legendre
- Numerical problems

Gauss-legendre Integration

Gauss-legendre integration is a numerical method used to evaluate definite integrals more accurately.

It is based on this form:

$$\int_{-1}^1 f(x) dx$$

Instead of using equally spaced points like trapezoidal or Simpson's rule, this method uses special points called Gauss points and special weights.

The formula is :

$$\int_{-1}^1 f(x) dx = \sum_{i=1}^n w_i f(x_i)$$

where,

w_i = weights

x_i = Gauss points

If limits are not $[-1, 1]$

then first change the variable,

$$\int_a^b f(x) dx$$

$$x = \frac{b-a}{2} u + \frac{a+b}{2}$$

$$dx = \frac{b-a}{2} du$$

Then apply Gauss formula.

2 point Gauss-Legendre formula :

$$\int_{-1}^1 f(u) du = w_1 f(u_1) + w_2 f(u_2)$$

Here, $w_1 = w_2 = 1$

$$\therefore \int_{-1}^1 f(u) du = f(u_1) + f(u_2)$$

where, $u_1 = -\frac{1}{\sqrt{3}}$ and $u_2 = \frac{1}{\sqrt{3}}$

3 point Gauss-Legendre formula :

$$\int_{-1}^1 f(u) du = w_1 f(u_1) + w_2 f(u_2) + w_3 f(u_3)$$

OR

$$\int_{-1}^1 f(u) du = \frac{5}{9} f\left(-\sqrt{\frac{3}{5}}\right) + \frac{8}{9} f(0) + \frac{5}{9} f\left(\sqrt{\frac{3}{5}}\right)$$

where, $w_1 = 0.555$

$$u_1 = -0.7740$$

$$w_2 = 0.889$$

$$u_2 = 0$$

$$w_3 = 0.555$$

$$u_3 = 0.7740$$

4 point Gauss-Legendre formula :

$$\int_{-1}^1 f(u) du = w_1 f(u_1) + w_2 f(u_2) + w_3 f(u_3) + w_4 f(u_4)$$

where,

$$w_1 = 0.34785$$

$$u_1 = -0.86144$$

$$w_2 = 0.65214$$

$$u_2 = -0.33998$$

$$w_3 = 0.65214$$

$$u_3 = 0.33998$$

$$w_4 = 0.34785$$

$$u_4 = 0.86144$$

Numerical Problem :

Q: Compute $\int_{-1}^1 e^x dx$ using two point Gauss-Legendre formula.

Solution: Here limit of integration is -1 and 1 so we can directly apply formula,

$$\int_{-1}^1 f(u) du = w_1 f(u_1) + w_2 f(u_2)$$

$$= f\left(-\frac{1}{\sqrt{3}}\right) + f\left(\frac{1}{\sqrt{3}}\right)$$

$$= e^{-1/\sqrt{3}} + e^{1/\sqrt{3}}$$

$$= 2.34269608 \text{ Ans.}$$

Q:2 Evaluate $\int_0^1 \frac{1}{1+x} dx$ using 2 & 3 point Gauss-Legendre method.

Solution: Transforming limit from $(0,1)$ to $(-1,1)$

we have,

$$x = \frac{b-a}{2}u + \frac{a+b}{2}$$

$$x = \frac{1-0}{2}u + \frac{0+1}{2}$$

$$x = \frac{u+1}{2}$$

$$dx = \frac{du}{2}$$

Now integration become, $\int_0^1 \frac{1}{1+x} dx,$

$$\int_{-1}^1 \frac{1}{1 + \frac{u+1}{2}} \times \frac{du}{2} = \int_{-1}^1 \frac{2}{u+3} \times \frac{du}{2} = \int_{-1}^1 \frac{du}{u+3}$$

Now applying Gauss-Legendre 2 point formula,

$$\int_{-1}^1 f(u) du = w_1 f(u_1) + w_2 f(u_2)$$
$$= f\left(-\frac{1}{\sqrt{3}}\right) + f\left(\frac{1}{\sqrt{3}}\right)$$

$$= \frac{1}{-\frac{1}{\sqrt{3}}+3} + \frac{1}{\frac{1}{\sqrt{3}}+3}$$

$$= 0.4127711 + 0.279536$$

$$= 0.6923076 \text{ Ans.}$$

Applying Gauss-Legendre 3 point formula,

$$\begin{aligned}\int_{-1}^1 f(u) du &= \frac{5}{9} f\left(-\sqrt{\frac{3}{5}}\right) + \frac{8}{9} f(0) + \frac{5}{9} f\left(\sqrt{\frac{3}{5}}\right) \\&= \frac{5}{9} \times \frac{1}{-\sqrt{\frac{3}{5}} + 3} + \frac{8}{9} \times \frac{1}{0 + 3} + \frac{5}{9} \times \frac{1}{\sqrt{\frac{3}{5}} + 3} \\&= 0.249642 + 0.2962962 + 0.1471827 \\&= 0.6931209 \text{ Ay.}\end{aligned}$$

Parameters for Gaussian Integration

n	i	w_i	z_i
2	1	1.00000	-0.57735
	2	1.00000	0.57735
3	1	0.55556	-0.77460
	2	0.88889	0.00000
	3	0.55556	+0.77460
4	1	0.34785	-0.86114
	2	0.65215	-0.33998
	3	0.65215	+0.33998
	4	0.34785	0.86114
5	1	0.23693	-0.90618
	2	0.47863	-0.53847
	3	0.56889	0.00000
	4	0.47863	0.53847
	5	0.23693	0.90618

(Contd.)

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